

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

3. In the graph $G = (V, E)$, with $V = \{0, 1, 2, 3, 4, 5\}$, and edges $((i, j), w)$, between i and j with distance w , in $E = \{((0, 3), 8), ((0, 1), 5), ((0, 2), 4), ((1, 4), 1), ((1, 2), 4), ((2, 3), 2), ((1, 3), 3), ((3, 4), 7), ((1, 5), 4), ((2, 5), 2), ((4, 5), 7)\}$, we want to compute all shortest paths from 0 to all other vertices.

- (a) Illustrate the execution of Dijkstra's algorithm on G .
Show all steps and the evolution of data structure to store the shortest paths.
- (b) Formulate the running time in function of the size of G . Justify your formula.

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Answer:

3. (a) S is the explored set, d are distances, p is path.
- i. $S = \{0, 2\}$, $d = [0, -1, 4, -1, -1, -1]$, $p = [0, -1, 0, -1, -1, -1]$
 - ii. $S = \{0, 2, 3\}$, $d = [0, -1, 4, 6, -1, -1]$, $p = [0, -1, 0, 2, -1, -1]$
 - iii. $S = \{0, 2, 3, 5\}$, $d = [0, -1, 4, 6, -1, 6]$, $p = [0, -1, 0, 2, -1, 2]$
 - iv. $S = \{0, 2, 3, 5, 1\}$, $d = [0, 5, 4, 6, -1, 6]$, $p = [0, 0, 0, 2, -1, 2]$
 - v. $S = \{0, 2, 3, 5, 1, 4\}$, $d = [0, 5, 4, 6, 6, 6]$, $p = [0, 0, 0, 2, 1, 2]$
- (b) For a graph with n vertices and m edges, the running time is $O(nm)$ as the outer loop runs over all n vertices and the inner loop considers all m edges.
- Using a priority queue to store the vertices with the distance to the first vertex 0, the running time is improved to $O(m \log(n))$.